



Fossil reveals ancient forest

Barneo's rainforests are 4 million years old, reveals a detailed study of leaf fossils.
<https://digit.in/may22-31>



Do batteries require visas too?

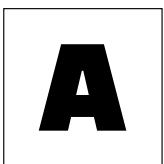
BMW takes part in research that will develop a battery passport to trace their carbon footprint.
<https://digit.in/may22-32>

Turning back time on skin ageing



This new method involves the reprogramming of old skin cells to bring back their youthful functionality

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Ageing can be looked at as the slow and gradual reduction in a living being's fitness that happens

due to the reduced functionality, altered gene expressions and perturbed epigenome of the cells of

the being. Over time this leads to tissue dysfunction and diseases that are considered age-related problems. This can also be seen in the cells as the genome accumulates marks of ageing. But if somehow those age-related markings in the genome could be erased, the cells could regain their youthful functions. This is exactly what the researchers over at Babraham Institute are aiming for with the help of regenerative biology.

But first, let's get a bit of a low-down on what exactly regenerative biology is all about.

One could say that the main aim of regenerative biology would be to either repair existing but damaged cells or replace them entirely with new ones. Now that can be done with the help of induced stem cells which is a process involving the erasure of some of the markings that give the cell's specialisation. The idea is that

these induced stem cells have, in theory, the potential to become any type of cell. So when these induced stem cells take on a particular specialisation, they can reverse the ageing process of whatever tissue they become a part of. Sadly though, scientists have not yet managed to reliably reprogramme the cells into specific specialisations. And that's where this new research from Babraham Institute comes in.

NEW METHODS OF REGENERATIVE BIOLOGY

So far we have established that scientists can completely erase a cell's identity to make them induced stem cells but cannot reliably reprogram them to a particular specialisation. There is a ray of hope though, thanks to a Nobel Prize-winning technique that halts the identity-erasing process partway so that the cells retain their specialised functions but behave like younger cells thanks to the age-related markings on them. So, how to achieve this partial erasure? Well, we need to go back to the beginning when Shinya Yamanaka became the first scientist to successfully turn normal cells into stem cells. His process took about 50 days and four molecules called the Yamanaka factors to make the cell completely lose its identity and become a stem cell. Adapting this process, a new method was created called the maturation Phase Transient Reprogram-